

Fact Sheet

US CO-ORBITAL ANTI-SATELLITE TESTING

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SUMMARY

The United States conducted one co-orbital intercept test in 1986 with the Delta 180 experiment as part of the Strategic Defense Initiative (SDI). The United States has conducted multiple tests of technologies for close approach and rendezvous in both low Earth Orbit (LEO) and Geostationary Orbit (GEO), along with tracking, targeting, and hit-to-kill (HTK) intercept technologies that could lead to a co-orbital anti-satellite (ASAT) capability. These tests and demonstrations were conducted for other non-offensive missions, such as missile defense, on-orbit inspections, and satellite servicing, and the United States does not have an acknowledged program to develop co-orbital capabilities. However, the United States possesses the technological capability to develop a co-orbital capability in a short period of time if it chooses to. If the United States moves ahead with its planned Golden Dome missile defense architecture, the space-based interceptors that would be part of such a program could provide a co-orbital capability.

CO-ORBITAL ASAT PROGRAMS

Co-orbital ASATs place an interceptor into orbit, which then maneuvers to alter its orbit to a trajectory that brings it close to a target. Co-orbital ASATs could maneuver to approach immediately after being placed into orbit or after remaining dormant for an extended period of time. They can try to damage or destroy their target by direct collision at hyper velocities, releasing a cloud of fragments that will collide with the target, using a robotic arm to damage or remove parts of a target satellite, or using electronic warfare or directed energy weapons at close range. Regardless of the technique used, co-orbital ASATs require onboard guidance, navigation, and control systems to identify and track a targeted space object and fine-tune its trajectory for proper interception. Although the United States has never had an officially recognized co-orbital ASAT program, it tested and developed many underlying technologies as part of its missile defense programs during the Cold War. Notably, several of the technologies for space-based ballistic missile intercept developed as part of the SDI during the 1980s could have been used to intercept satellites as well.

Project SAINT

Project SAINT (also known as the Satellite Inspector Program) was a US Air Force (USAF) effort to develop a system that could be used initially as a satellite inspector but eventually could be turned into a co-orbital ASAT weapon. The USAF developed initial ideas for three different concepts: one that was uncrewed and ground-launched, one that was uncrewed and air-launched, and a third that was crewed.¹ In 1960, the USAF pressed forward with a "satellite inspector" version of the program.² The inspector concept called for the SAINT vehicle to be launched into orbit on an Atlas booster, after which it would match orbits with the target and use onboard television cameras and radars to inspect the target from as close as 50 feet. However, the USAF also hoped that a later version of the SAINT vehicle would include a kill mechanism, such as high-explosive rockets. The USAF planned for an initial set of four intercept tests beginning in 1963 and SAINT to be fully operational by the summer of 1967.³ However, lack of budget support and political concerns led to the program's cancellation in 1962 before any on-orbit tests were

conducted.

Delta 180 Experiment

The United States conducted a successful co-orbital intercept as part of the Delta 180 experiment as part of its SDI. The goal of the Delta 180 experiment was to better understand tracking, guidance, and control for a space intercept of an accelerating target.⁴ The experiment involved modifying the second stage of a Delta 2 (D2) rocket to carry a sophisticated tracking system that included radar, ultraviolet, visible, and infrared sensors. The payload consisted of a McDonnell Douglas Payload Assist System (PAS) platform combined with the warhead and seeker from a Phoenix air-to-air missile and Delta 2 rocket motors.

The Delta 180 experiment was launched from the Cape Canaveral Air Force Station on September 5, 1986. Two pieces, the D2 target and the PAS interceptor, were placed into a 220-km circular orbit. The PAS maneuvered to a separation distance of 200 km, and 90 minutes after launch, the D2 observed the launch of an Aries rocket from White Sands Missile Range. At 205 minutes after launch, the D2 and PAS both ignited their engines on an intercept course, colliding at a combined speed of nearly 3 km/s.⁵ Sixteen pieces of orbital debris from the collision were cataloged with apogees as high as 2,300 km. However, the low altitude of the intercept resulted in most of pieces reentering the atmosphere within two months, with all of them having deorbited within seven months.⁶

CURRENT STATUS

While the United States has not had a fully structured co-orbital ASAT program, it has tested and developed many of the underlying technologies. These include multiple successful military robotic rendezvous and proximity operations (RPO), between satellites in both LEO and GEO, of sophisticated space situational awareness (SSA) capabilities for identifying, tracking, and targeting space objects. Examples of RPO missions include the 1990 Prowler satellite that maneuvered around the GEO region to inspect Russian satellites,⁷ the 2006 XSS-11 satellite that conducted RPO and inspections of LEO space objects,⁸ and the GEO Space Situational Awareness Program (GSSAP) satellites that are currently on orbit conducting inspections and surveillance in the GEO region.⁹

In January 2025, President Donald Trump released an Executive Order (EO) called “An Iron Dome for America, later renamed as “Golden Dome for America,” with the purpose of creating a next-generation missile defense shield.”¹⁰ Golden Dome is intended to have a space-based interceptor layer; if the United States does test and field these interceptors, they could provide a co-orbital counterspace capability.

Historical US Co-Orbital ASAT Tests in Space

Date	Interceptor	Launch Site	Target	Orbit Altitude	Debris Created ¹¹	Result
Sep. 1986	Delta 180 Payload Adapter System	Cape Canaveral Air Force Station	Delta 2 R/B	220 km	16	The successful intercept of thrusting object in a 220-km circular orbit

ENDNOTES

1. Paul Stares, *The Militarization of Space: U.S. Policy, 1945-1984*, Cornell University Press, August 1, 1985, pg. 112.
2. Stares, *The Militarization of Space: U.S. Policy, 1945-1984*, *ibid*, pg. 112-113.
3. Stares, *The Militarization of Space: U.S. Policy, 1945-1984*, *ibid*, pg. 115.
4. John Dassoulas and Michael D. Griffin, "The Creation of the Delta 180 Program and Its Follow-ons," *Johns Hopkins APL Technical Digest*, vol. 11, Numbers 1 and 2 (1990): pg. 86, <https://www.jhuapl.edu/Content/techdigest/pdf/V11-N1-2/11-01-Dassoulas.pdf>.
5. "VSE (Delta-180, DM-43)," *Gunter's Space Page*, accessed March 22, 2018, https://space.skyrocket.de/doc_sdat/vse.htm.
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7. Ted Molczan, "Unknown GEO Object 2000-653A / 90007 identified as Prowler," January 21, 2011, https://satobs.org/seesat_ref/STS_38/Unknown_GEO_Object_2000-653A_90007_Identified_as_Prowler.pdf.
8. "XSS-11 Micro Satellite," Air Force Research Laboratory, Space Vehicles Directorate, current as of September 2011, accessed March 22, 2018, p.1, <http://www.kirtland.af.mil/Portals/52/documents/AFD-111103-035.pdf?ver=2016-06-28-110256-797>.
9. Stephen Clark, "Air Force General Reveals New Space Surveillance Program," *SpaceFlight Now*, February 25, 2014, <http://spaceflightnow.com/news/n1402/25spasat/>; Colin, Clark, "US, China, Russia Test New Space War Tactics: Sats Buzzing, Spoofing, Spying," *Breaking Defense*, October 28, 2021, <https://breakingdefense.com/2021/10/us-china-russia-test-new-space-war-tactics-satsbuzzing-spoofing-spying/>.
10. The White House, "The Iron Dome for America," January 27, 2025, <https://www.whitehouse.gov/presidential-actions/2025/01/the-iron-dome-for-america/>.
11. Data compiled from the public catalog maintained by the US military at <https://Space-Track.org>.



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